Experiment Title: Viewing and Editing Files in Linux Terminal

Topic: File Viewing Tools (`cat`, `less`, `head`, `tail`) and Text Editors (`nano`, `vim`)

1. Objective

To practice terminal file viewers and text editors:

- View file contents (**`cat`**, **`head`**, **`tail`**, **`less`**).
- Check and install editor packages (**`which`**, **`sudo apt install`**).
- Edit text with **`nano`** (save, exit, cut, paste, find, replace).
- Edit text with **`vim`** (modes, yank, delete, put, search, substitute).

2. Theoretical Background

Linux uses viewers (pagers) for reading files and command-line text editors for modifying files without a GUI.

2.1 File Viewing Commands

Command	Syntax	Simple Meaning / Purpose	Example
`cat`	`cat <file>`	Print entire file content.	`cat config.txt`
`less`	`less <file>`	View file page by page (scrollable).	`less system.log`
`head`	`head -n <N> <file>`	Show first N lines.	`head -n 5 sample.txt`
`tail`	`tail -n <N> <file>`	Show last N lines (recent logs).	`tail -n 3 server.log`

2.2 Comparing Nano and Vim

Editor	Design Style	Best Used For
GNU Nano	Modeless editor with onscreen shortcuts (^ for Ctrl, M- for Alt).	Quick edits, configuration changes, beginners.
Vim	Modal editor with Normal, Insert, and Command modes.	Fast navigation, script editing, advanced users.

3. Equipment / Software Requirements

Item	Details	Item	Details
Operating System	Linux (Ubuntu 24.04 LTS)	**Host Machine**	`firefly`
User Account	`ratul` (with sudo)	**Terminal Shell**	GNU Bash 5.2
Viewing Utilities	GNU Coreutils (`cat`, `head`, `tail`), `less`	**Text Editors**	GNU Nano 7.2, Vim 9.1

4. Procedure / Steps

Steps followed in this experiment:

1. Display full log with **`cat server.log`**.
2. View first 3 lines with **`head -n 3`** and last 3 lines with **`tail -n 3`**.
3. View interactively with **`less server.log`** and exit with **q**.
4. Verify and edit with nano: search, replace, cut, paste, save (**Ctrl+O**), and exit (**Ctrl+X**).
5. Verify and edit with vim: switch modes (**i**, **Esc**), copy (**yy**), paste (**p**), cut (**dd**), substitute (**`:%s`**), and save-quit (**`:wq`**).

5. Result / Output Demonstration

All commands were executed under user `ratul` on host machine `firefly`.

5.1 Part 1: Viewing Files with `cat`, `head`, `tail`, and `less`

ratul@firefly: ~/lab4 — bash

```
ratul@firefly:~/lab4$ cat server.log
[2026-10-08 09:00:01] INFO System startup complete
[2026-10-08 09:05:12] DEBUG Loading database connection pool
[2026-10-08 09:12:44] WARN High CPU usage detected on core 2
[2026-10-08 09:20:10] ERROR Port 8080 timeout connecting to worker
[2026-10-08 09:30:00] INFO Scheduled backup process finished

ratul@firefly:~/lab4$ head -n 2 server.log
[2026-10-08 09:00:01] INFO System startup complete
[2026-10-08 09:05:12] DEBUG Loading database connection pool

ratul@firefly:~/lab4$ tail -n 2 server.log
[2026-10-08 09:20:10] ERROR Port 8080 timeout connecting to worker
[2026-10-08 09:30:00] INFO Scheduled backup process finished

ratul@firefly:~/lab4$ less server.log
server.log (lines 1-5/5) (Press q to quit, / to search)
```

Explanation of Part 1 Results:

- `cat`: Printed entire file to terminal.
- `head` and `tail`: Extracted top 2 lines and bottom 2 lines of log.
- `less`: Opened scrollable view without changing file contents.

5.2 Part 2: Installing and Editing with GNU Nano

ratul@firefly: ~/lab4 — bash

```
ratul@firefly:~/lab4$ which nano
/usr/bin/nano

ratul@firefly:~/lab4$ sudo apt install nano
nano is already the newest version (7.2-1ubuntu1).

ratul@firefly:~/lab4$ nano app_config.txt
GNU nano 7.2                File: app_config.txt                Modified
server_mode = production
max_connections = 500
port = 9000
[ Replace "development" with "production" completed: 1 replacement ]
^G Get Help  ^O WriteOut  ^W Where Is  ^K Cut     M-U Undo     ^X Exit
^R Read File ^\ Replace   ^_ Go To Line ^U Paste   M-E Redo     M-6 Copy

ratul@firefly:~/lab4$ cat app_config.txt
server_mode = production | max_connections = 500 | port = 9000
```

Explanation of Part 2 Results:

- Verified nano with `which nano` and `sudo apt install nano`.
- Replaced text with **Ctrl+**, saved with **Ctrl+O**, and exited with **Ctrl+X**.

5.3 Part 3: Installing and Editing with Vim

ratul@firefly: ~/lab4 — bash

```
ratul@firefly:~/lab4$ which vim
/usr/bin/vim
ratul@firefly:~/lab4$ sudo apt install vim
vim is already the newest version (2:9.1.0-1ubuntu1).
ratul@firefly:~/lab4$ vim script.sh
~                    VIM - Vi IMproved 9.1
1 #!/usr/bin/env bash
2 echo "Starting backup service..."
3 echo "Status: service running successfully."
4 echo "Completed backup."
- INSERT -                                          4,1       All
:%s/task/service/g   2 substitutions on 2 lines
:wq                  "script.sh" 4L, 114B written
ratul@firefly:~/lab4$ cat script.sh
#!/usr/bin/env bash
echo "Starting backup service..."
echo "Status: service running successfully."
echo "Completed backup."
ratul@firefly:~/lab4$ _
```

Explanation of Part 3 Results:

- Switched between Normal mode and Insert mode (**i**, **Esc**).
- Performed copy (**yy**), paste (**p**), cut (**dd**), and undo/redo (**u**, **Ctrl+r**).
- Replaced text with **`:%s/task/service/g`** and saved with **`:wq`**.

6. Conclusion

Key takeaways from this lab:

1. Inspected files using **`cat`**, **`head`**, **`tail`**, and **`less`**.
2. Edited text with simple shortcuts in GNU Nano.
3. Edited text with modal commands, copy/paste, and search/replace in Vim.